# Lam Research Challenge 3.0 — Stage 1

Worked Solutions: Question 2.4, and Task 3 (Q3.1 – Q3.5) — with Supporting Background Theory

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Given process parameters used throughout: pipe diameter = 10 mm, fluid density $\rho$ = 1000 kg/$m^3$, dynamic viscosity $\mu$ = $1\times10^{-3}$ Pa·s, flow range 1–6 L/min, max valve Delta-P = 1.5 bar, SG = 1, pressure transmitter range 0–4 bar, ADC = 12-bit, sensor output 4–20 mA, supply = 24 VDC, Delta-T = 35°C, pipeline length = 3 m, Darcy friction factor f = 0.03, plant transfer function G(s) = 6/(4s+1), Cp = 4.18 kJ/kg·K, U = 500 W/$m^2$K, R (air) = 0.287 kPa·$m^3$/(kg·K), atmospheric pressure = 1.013 bar.

# PART A — WORKED SOLUTIONS

## Q2.4 — Heat Exchanger Area (LMTD Method)

**Given:** Process fluid heated 25°C → 60°C; hot water 80°C → 65°C; max flow = 6 L/min; U = 500 W/m$^2$K. Counter-current arrangement is assumed (standard default for LMTD sizing unless co-current is specified, and it gives the smaller/realistic required area).

### *Step 1 — Heat duty (Q) required by the process fluid*

```
Q = ṁ × Cp × Delta-T_process (ṁ = mass flow rate)
```

Volumetric flow: 6 L/min = 6×10$^{-3}$ m$^3$ / 60 s = 1×10$^{-4}$ m$^3$/s
Mass flow: ṁ = ρ × Q = 1000 × 1×10$^{-4}$ = 0.1 kg/s
Delta-T_process = 60 – 25 = 35°C

```
Q = 0.1 kg/s × 4180 J/kg·K × 35 K = 14 630 W ≈ 14.63 kW
```

### *Step 2 — Log Mean Temperature Difference (counter-current)*

Hot-in pairs with cold-out; hot-out pairs with cold-in:
Delta-T$_1$ = T_hot,in – T_cold,out = 80 – 60 = 20°C
Delta-T$_2$ = T_hot,out – T_cold,in = 65 – 25 = 40°C

```
LMTD = (Delta-T2 – Delta-T1) / ln(Delta-T2/Delta-T1) = (40–20) / ln(40/20) =
20 / 0.693 = 28.85°C
```

### *Step 3 — Required heat-transfer area*

```
A = Q / (U × LMTD)
A = 14 630 / (500 × 28.85) = 14 630 / 14 425 = 1.014 m²
```

### *Step 4 — Effect of 20% fouling*

Fouling adds a resistance in series with the clean overall coefficient: 1/U_dirty = 1/U_clean + R_f. Taking the stated 20% degradation as a 20% drop in effective U (U_dirty = 0.8 × 500 = 400 W/m$^2$K), and Q and LMTD unchanged:

```
A_dirty = 14 630 / (400 × 28.85) = 1.268 m² (≈ 25% larger than the
clean-surface area)
```

**Conclusion:** Fouling reduces the overall heat-transfer coefficient, so a larger heat-transfer area (or equivalently a higher installed design margin) is needed to deliver the same duty. If the physical exchanger area is fixed at 1.014 m$^2$, fouling instead causes the outlet temperatures to fall short of the 60°C target (under-performance) unless compensated by increasing flow, temperature, or cleaning frequency.

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## Q3.1 — Pressure Sensor Resolution

**Given:** Pressure range 0–4 bar, ADC = 12-bit (2$^{12}$ = 4096 discrete levels).

```
Resolution = Full-Scale Span / 2^n
Resolution = 4 bar / 4096 = 0.000977 bar ≈ 0.98 mbar per bit (≈ 0.0244% of
span per LSB)
```

**Sufficiency check for ±1% control:** ±1% of a 4 bar span = ±0.04 bar (40 mbar) of allowable measurement error/step. The ADC's quantisation step (≈0.98 mbar) is about 41× finer than the 40 mbar tolerance band. **Yes — the 12-bit ADC resolution is more than sufficient for ±1% control**; the limiting factor for overall loop accuracy in practice will be sensor accuracy/linearity and noise, not ADC quantisation.

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## Q3.2 — Temperature Sensor Calibration Graph (4–20 mA, 0–100°C)

```
Linear transfer function: T = [(I − 4) / (20 − 4)] × 100 = 6.25 × (I − 4)
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| Current (mA) | 4 | 8 | 12 | 16 | 20 |
|---|---|---|---|---|---|
| Temperature (°C) | 0 | 25 | 50 | 75 | 100 |

The calibration curve is a straight line from (4 mA, 0°C) to (20 mA, 100°C), slope = 6.25 °C/mA, passing through the points above.

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At I = 12 mA: T = 6.25 × (12−4) = 50°C (the electrical mid-scale point
correctly maps to the physical mid-scale temperature, confirming linearity).
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## Q3.3 — 4–20 mA to 0–3.3 V Signal-Conditioning Circuit

**Approach:** A precision shunt (burden) resistor converts the loop current to a voltage via Ohm's Law, sized so the maximum loop current (20 mA) produces the ADC's full-scale voltage (3.3 V).

```
R_shunt = V_FS / I_FS = 3.3 V / 0.020 A = 165 Ω
```

Checking the low end: at 4 mA (live-zero), V = 0.004 × 165 = 0.66 V — not 0 V. This is expected and correct behaviour of a 4–20 mA "live zero" loop (as opposed to 0–20 mA). If the microcontroller/PLC needs a true 0–3.3 V signal representing 0–100% of range, an instrumentation-amplifier stage (or a software offset/scaling correction: %span = (V – 0.66)/(3.3–0.66) × 100) is added after the shunt to remove the 0.66 V offset. A low-pass RC filter (e.g. R ≈ 1 kΩ, C ≈ 0.1 µF) and a clamping/protection diode to the supply rail are typically added ahead of the ADC pin for noise rejection and over-voltage protection.

***Why signal conditioning is needed between transmitter and controller:***

- The 4–20 mA current loop is chosen for transmission because current, unlike voltage, does not attenuate with cable resistance/length, giving strong noise immunity over long industrial cable runs — but the controller's ADC/microcontroller input only accepts voltage.
- Signal conditioning performs current-to-voltage conversion, level shifting/scaling into the ADC's exact input window (0–3.3 V here), filtering of induced EMI/noise, and protection (clamping, isolation) of the sensitive digital input from field-side transients, fault currents or miswiring.

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## Q3.4 — Sensor Drift Correction (+0.08 bar after 120 days)

> Drift rate observed ≈ 0.08 bar / 120 days ≈ 0.00067 bar/day — useful for setting the next calibration interval so that error stays inside tolerance between calibrations.

**(a) SOP:** Isolate the transmitter and vent it to atmosphere; record the as-found zero (+0.08 bar). Check span at 0, 25, 50, 75, 100% of the 0–4 bar range against a traceable reference (dead-weight tester). Adjust zero/span, record as-left values, and issue a calibration certificate.

**(b) Zero correction:** Apply a –0.08 bar zero-trim offset (digital trim on smart transmitters, or a physical screw on analogue units); corrected reading = raw reading – 0.08 bar across the range.

**(c) Field validation:** Re-check the corrected transmitter at 0, 50, 100% of span against a portable reference, confirm the DCS/PLC shows the corrected value end-to-end, and set the next due-date from the drift rate.

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## Q3.5 — Unstable SOV Outlet Pressure with Coil Heating

**Given:** ½-inch line at 15 bar; Aira Euro solenoid valve (SOV), 24 VDC, 8 W coil; SOV rated 20 bar / 150°C max. Outlet pressure unstable and coil overheats.

Rated coil current: I = P/V = 8/24 = 0.333 A  Coil resistance: R = V/I = 24/0.333 = **72** Ω (check: R = $V^2$/P = 576/8 = 72 Ω). Since 15 bar is below the 20 bar rating, the SOV is not over-pressured — the fault lies elsewhere. Structured diagnosis and remedy:

### *Case 1 — Coil electrical fault / duty-cycle overload*

Measure coil current/resistance against the 72 Ω / 0.333 A rating. A shorted turn lowers resistance and raises current and heat; a coil rated for intermittent duty but energised continuously will also overheat, as will sustained over-voltage above 24 VDC (raises $I^2R$ heating). **Remedy:** verify supply voltage, confirm duty-cycle rating matches usage, replace coil if resistance is off-spec.

### *Case 2 — Mechanical/flow-path fault causing chatter*

A worn seat, debris in the orifice, or a damaged diaphragm/plunger can prevent full seating, causing rapid open/close chatter — destabilising outlet pressure and raising average coil current via repeated inrush. **Remedy:** inspect valve internals, clean or replace the seat/diaphragm/plunger, fit an upstream strainer if debris is the cause.

### *Case 3 — Insufficient minimum operating pressure differential*

Pilot-operated valves need a minimum inlet/outlet Delta-P to fully open the main poppet; below the manufacturer's minimum, the valve may hunt between partially open/closed states, giving unstable outlet pressure. **Remedy:** check operating Delta-P against the datasheet's minimum and adjust the upstream setup if too low.

### *Case 4 — Poor heat dissipation / ambient conditions*

Even at rated power, a coil in a poorly ventilated or hot enclosure can run close to its 150°C limit. **Remedy:** improve ventilation/spacing, add a heat sink, or relocate the SOV away from other heat sources; monitor coil temperature and de-rate/replace if needed.

**Overall recommendation:** with 15 bar well within the 20 bar rating, first rule out electrical causes (Case 1), then mechanical chatter (Case 2), then minimum-Delta-P conditions (Case 3), and finally thermal/installation environment (Case 4).

# PART B — BACKGROUND THEORY REQUIRED

The theory needed to solve Q2.4 and Q3.1–3.5 spans heat exchanger sizing, ADC/instrumentation resolution, current-loop signal transmission and conditioning, calibration/metrology practice, and solenoid-valve/electromagnetic-actuator fundamentals. Each topic below gives the actual working relations, not just the name of the concept.

## 1. Heat Exchanger Sizing — LMTD Method (needed for Q2.4)

**Sensible heat duty.** The rate of heat that must be transferred to change a fluid's temperature (no phase change) is:

$$Q = \dot{m} \times Cp \times \Delta T$$

where $\dot{m}$ is mass flow rate (kg/s), Cp is specific heat capacity (J/kg·K), and Delta-T is the temperature rise/fall of that stream. Mass flow is obtained from volumetric flow via $\dot{m} = \rho \times Q_{vol}$.

**Overall heat-transfer rate equation.** For a heat exchanger, the same duty Q must also be transferred across the wall separating hot and cold streams:

$$Q = U \times A \times \Delta T_{lm}$$

where U is the overall heat-transfer coefficient ($W/m^2K$, combining both film coefficients and wall/fouling resistances), A is the heat-transfer area, and Delta-T_lm is the Log Mean Temperature Difference — the correct "average" driving temperature difference when it varies along the exchanger length.

**LMTD formula.** For two terminal temperature differences $\Delta T_1$ and $\Delta T_2$ (one at each end of the exchanger):

$$\Delta T_{lm} = (\Delta T_2 - \Delta T_1) / \ln(\Delta T_2/\Delta T_1)$$

**Counter-current vs co-current flow.** In counter-current (streams flow in opposite directions), the hot-inlet is matched against the cold-outlet and hot-outlet against cold-inlet, which generally gives a larger LMTD (and hence a smaller required area) than co-current (parallel) flow for the same terminal temperatures — this is why counter-current is the standard default assumption and preferred industrial arrangement.

**Fouling.** Deposits (scale, biological growth, particulates) that build up on heat-transfer surfaces over time add an extra thermal resistance in series with the clean-surface resistance:

$$1/U_{dirty} = 1/U_{clean} + R_f$$

where R_f is the fouling resistance ($m^2K/W$). Since resistances add, U_dirty < U_clean, so for a fixed duty Q and LMTD, a larger area is needed to compensate ($A \propto 1/U$ from $Q=UA\cdot\Delta T_{lm}$) — or, for a fixed installed area, the achievable duty/outlet temperature falls short of design.

## 2. ADC Resolution and Quantisation (needed for Q3.1)

An analogue-to-digital converter (ADC) divides a continuous input span into $2^n$ discrete digital codes, where n is the number of bits. The smallest change in the physical quantity that changes the digital output by one count (one LSB, least-significant bit) is the resolution:

$$\text{Resolution} = \text{Full-Scale Span} / 2^n \text{ (or, more precisely, Span}/(2^n - 1))$$

This quantisation introduces an inherent uncertainty of up to ±½ LSB in every digitised reading (quantisation error), separate from the sensor's own accuracy/linearity/noise specifications. To judge whether a given ADC is "good enough" for a required control tolerance (e.g. ±1% of span), the resolution is compared against the absolute tolerance band; the ADC resolution should be a small fraction (rule of thumb: ≤1/10th, ideally much less) of that tolerance so that quantisation is not the limiting error source.

## 3. 4–20 mA Current Loop and Signal Conditioning (needed for Q3.2, Q3.3)

**Why current loops are used.** Industrial field transmitters commonly output an analogue signal as a current (not voltage) between 4 mA and 20 mA, proportional to the measured variable across its calibrated range. Current signals are immune to voltage drop along long cable runs and largely immune to induced electrical noise (because the same current flows through every point of the series loop regardless of cable resistance), which voltage signals are not.

**"Live zero".** The lower end of the range is defined as 4 mA (not 0 mA) rather than a "dead zero". This lets the receiving system distinguish a genuine 0%-of-range reading (4 mA, wire intact) from a broken wire or powered-off transmitter (0 mA) — an inherent open-circuit/fault detection feature.

**Linear scaling (current-to-engineering-units conversion).** Given a sensor's engineering-unit range [Lo, Hi] mapped linearly onto 4–20 mA:

$$Value = Lo + [(I - 4)/(20 - 4)] \times (Hi - Lo)$$

**Current-to-voltage conversion (signal conditioning).** Since digital controllers/ADC pins read voltage, not current, a precision shunt (burden) resistor is placed across the loop, converting current to a proportional voltage by Ohm's Law:

$$V = I \times R_{shunt}$$

The shunt value is chosen so the maximum loop current (20 mA) produces the ADC's full-scale voltage. Because the loop's zero is "live" at 4 mA rather than 0 mA, the resulting voltage at the low end is not 0 V; additional conditioning (instrumentation amplifier for offset removal/scaling, RC low-pass filtering for noise, and clamping/protection diodes) is typically added between the shunt and the ADC input to present a clean, correctly-scaled, protected 0–V_FS signal.

## 4. Instrument Calibration and Drift (needed for Q3.4)

**Zero drift vs span drift.** Over time, an instrument's output can drift due to component ageing, temperature cycling, mechanical stress or contamination. A *zero drift* is a constant offset added across the whole range (the reading at the true-zero input point moves away from zero); a *span drift* is a change in the slope/gain of the output vs input relationship (error grows with reading). Distinguishing the two requires checking more than one point (at minimum zero and full-scale, ideally several intermediate points).

**Calibration hierarchy/traceability.** Field instruments are checked against a reference standard (e.g. a dead-weight tester for pressure, a calibrated portable gauge) whose own accuracy is traceable to a recognised national/international standard, ensuring the correction applied is itself trustworthy.

**As-found / as-left recording.** Good calibration practice always records the reading *before* adjustment (as-found — used to judge whether the instrument was within tolerance during the preceding service period) and *after* adjustment (as-left — the corrected baseline for the next period), which also allows drift rate to be tracked and used to set the next calibration interval.

## 5. Solenoid Valve (SOV) Operating Principles (needed for Q3.5)

**Basic operation.** A solenoid valve uses an electromagnetic coil to move a plunger/armature that opens or closes a flow path (directly, in a direct-acting valve, or by opening a small pilot orifice that lets process pressure move a larger main poppet, in a pilot-operated valve).

**Coil electrical relations.** For a DC coil, rated power, voltage and current are related by Ohm's Law and the power equation:

$$P = V \times I = I^2R = V^2/R$$

Excess current (from over-voltage, a partially shorted winding, or duty-cycle beyond rating) raises $I^2R$ heating in the coil.

**Minimum operating pressure differential (pilot-operated valves).** Pilot-operated solenoid valves rely on a minimum pressure differential across the valve to fully lift the main poppet using process pressure itself (the small solenoid only actuates the pilot). Operating below the manufacturer's stated minimum Delta-P can prevent full, stable opening, causing hunting/chatter and pressure instability at the outlet.

**Duty cycle and thermal rating.** Solenoid coils are rated for either continuous duty (100% ED, safe to energise indefinitely) or intermittent duty (a maximum "on" fraction per cycle). Energising an intermittent-duty coil continuously, or exceeding the coil's maximum ambient/case temperature rating, leads to overheating and accelerated insulation/winding failure — this is checked against the manufacturer's maximum coil temperature and duty-cycle specification.